

6.1 The Basics of Counting

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How many passwords can be made with exactly 12 characters which may be only lower case letters a-z or numbers 0-9?

The Sum Rule: If a task can be done in n_1 ways, or n_2 different ways, or n_3 different ways, ..., or n_k different ways, then there are $n_1 + n_2 + \dots + n_k$ ways to complete the task

Product Rule: If a procedure can be broken down into k independent tasks $T_1, T_2, \dots, T_k$ and if there are n_1 ways to complete T_1 , n_2 ways to complete T_2 , ..., n_k ways to complete T_k , then there are $n_1 \cdot n_2 \cdot \dots \cdot n_k$ ways to complete the procedure.

Password _ _ _ _ _

Each of the 12 characters are independently chosen. Each choice gives a "task".

Any character can be a letter a-z or a number 0-9, so there are $26 + 10$ choices for each character by the

Sum rule (since letters are different from numbers). Then there are 36 ways to complete each of the 12

tasks. Then there are

$$(36)(36)(36)(36)(36)(36)(36)(36)(36)(36)(36)(36) \\ = (36)^{12} \approx 4.74 \times 10^{18}$$

different passwords by the Product Rule

What if a task can be done n ways or m ways but some of them overlap?

Subtraction Rule: If a task can be done n ways or m ways, then the number of ways the task can be completed is $n + m - (\# \text{ ways they have in common})$.

This is also called the inclusion-exclusion principle

How many bit strings with 6 digits either start with 0 or end with 11?

$$\underline{0} \text{ --- } \leftarrow 2^5 \text{ ways} \\ 2 \cdot 2 \cdot 2 \cdot 2 \cdot 2$$

$$\text{---} \underline{11} \leftarrow 2^4 \text{ ways} \\ 2 \cdot 2 \cdot 2 \cdot 2$$

$$\underline{0} \text{ ---} \underline{11} \leftarrow 2^3 \text{ ways (overlap)} \\ 2 \cdot 2 \cdot 2$$

By the Subtraction Rule (and Product Rule)

there are $2^5 + 2^4 - 2^3$ many bit strings

$$2^5 + 2^4 - 2^3 = 2^3(2^2 + 2 - 1) = 2^3(5) = 40$$

A new disc golf company has 12 different disc models called molds. Each mold comes in 4 different colors and can be bought in weights 165g, 168g, 170g, and 173g. How many different types of disc are offered?

Any disc has 3 independent attributes

Any disc has 3 independent attributes

<u>mold</u>		<u>color</u>		<u>weight</u>
12	·	4	·	4

Using the Product Rule, there are $12 \cdot 4 \cdot 4 = 192$ different discs.

How many different three-letter initials can people have? How many of these with no repeated letters?

$_ _ _ \rightarrow 26^3$ three-letter initials
 $26 \cdot 26 \cdot 26$
any A-Z

$_ _ _ \rightarrow 26 \cdot 25 \cdot 24$ three-letter initials, no repeats
 $26 \cdot 25 \cdot 24$
any A-Z any A-Z except one any A-Z except 2

How many 7-digit numbers are palindromes?

$_ _ _ _ _ _ _$
 $\downarrow \downarrow \downarrow \downarrow \downarrow \downarrow \downarrow$
 $9 \quad 10 \quad 10 \quad 10 \quad _ \quad _ \quad _$
 $\downarrow \quad \quad \downarrow \downarrow \downarrow$
Any 1, ..., 9 Any 0, ..., 9

So there are $9 \cdot 10 \cdot 10 \cdot 10 = 9000$ 7-digit palindromes.

How many 7-digit numbers are not palindromes?

$_ _ _ _ _ _ _ \quad 9 \cdot 10^6 = 9000000$
 $9 \cdot 10 \cdot 10 \cdot 10 \cdot 10 \cdot 10 \cdot 10$
7-digit numbers.

Since there are 9000 7-digit palindromes,
there are $9000000 - 9000$ 7-digit numbers
that are not palindromes.